SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Problem-Solution Fit

Date	27 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS]

- Disaster Management Authorities
- Government Agencies
- Local Municipal Corporations
- Residents of Flood-Prone Areas
- Farmers
- Environmental Organizations

6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL]

- Limited technical knowledge
- Limited access to real-time data
- Internet connectivity issues in rural areas
- Dependence on manual reports
- Low awareness about early flood prediction tools

5. AVAILABLE SOLUTIONS PROS & CONS [AS]

Existing Solutions:

- Weather forecast websites
- Government flood alerts
- Manual observation & reports
- Traditional forecasting methods

Pros

- Accessible
- Official updates
- Basic forecasts

Cons

- Delayed alerts
- Low accuracy
- No personalization
- No ML-based analysis

2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- Delayed flood warnings
- Inaccurate flood prediction
- Heavy dependence on manual analysis
- Lack of centralized system
- Floods cause loss of life and property frequently

9. PROBLEM ROOT / CAUSE [RC]

- Heavy rainfall & extreme weather
- Rising river water levels
- Climate change
- Poor drainage & infrastructure
- Unstructured & scattered data
- Lack of predictive analytics & early warning systems

7. BEHAVIOR + ITS INTENSITY [BE]

- Frequently checks weather updates
- Searches for flood-related news
- Follows government advisories
- Takes action only after official alerts
- High activity during monsoon season (High intensity)

3. TRIGGERS TO ACT [TR]

- Heavy rainfall warnings
- Rising water levels in rivers/lakes
- Sudden weather changes
- Government alerts & advisories
- Need for disaster preparedness

4. EMOTIONS BEFORE / AFTER [EM]

Before:
Worried • Uncertain • Anxious • Unprepared

After:
Confident • Informed • Prepared • Safer

10. YOUR SOLUTION [SL]

- ML-based flood prediction system
- Uses historical data & ML models (XGBoost, RF, etc.) for accuracy
- Real-time input of environmental parameters
- Secure login & role-based access
- Instant predictions with probability score
- Prediction history & reports
- Dashboard with insights & analytics
- Early warning support for better decision-making

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Web application (Flood Prediction System)
- Government weather portals
- Social media updates
- News websites & mobile browsers

OFFLINE

- Disaster management offices
- Local government announcements
- TV / Radio / Newspapers
- Community meetings & awareness programs